NETWORK SEGMENTATION

Departmental Subnetting & Isolated Network Design

Simulated in VirtualBox | 15 Virtual Machines | 5 Departmental Subnets | NATNetwork Isolation

Base Network	`192.168.0.0/24`
Subnetting Scheme	8x /27 subnets (30 usable hosts per subnet),+3 Excess of req. subnets, suitable for future expansion
Departments Segmented	IT, Procurement, Audit, Sales and Guest
Virtualisation Platform	Oracle VirtualBox — NATNetwork feature
Total VMs Deployed	15 (2 Employee workstations + 1 Server per department)
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1. Executive Summary

This document presents the technical design and implementation of a departmental network segmentation project, simulated entirely within an Oracle VirtualBox lab environment. The project demonstrates the application of subnetting (CIDR/VLSM) principles to divide a single network block into five logically and technically isolated departmental subnets: Information Technology (IT), Procurement, Audit, Guest and Sales.

The base network 192.168.0.0/24 was subdivided into five equal-sized /27 subnets, each providing 30 usable host addresses — comfortably accommodating the lab's per-department device count while leaving headroom for future growth. Network isolation was implemented using VirtualBox's NATNetwork feature, creating five independent virtual networks, each acting as the equivalent of a separate VLAN or physical network segment in a production environment.

Fifteen virtual machines were deployed across the five subnets — two employee workstation VMs and one server VM per department — for a total of 10 workstations and 5 servers. Each VM's network adapter was explicitly attached to its corresponding departmental NATNetwork, ensuring that devices in one department cannot directly communicate with devices in another without passing through a controlled routing or firewall boundary. This project demonstrates core network security principles including the reduction of lateral movement risk, the containment of a potential breach to a single department, and the enforcement of the principle of least privilege at the network layer.

5	/27	15	30
Departmental Subnets	Subnet Mask Per Department	Virtual Machines Deployed	Usable Hosts Per Subnet

2. Project Objectives

2.1 Primary Objectives

- Design a subnetting scheme that divides a single /27 network block into five equally-sized departmental subnets using CIDR notation
- Simulate a realistic enterprise network topology within VirtualBox, including dedicated server and workstation roles per department
- Implement network-layer isolation between departments using VirtualBox's NATNetwork feature, replicating VLAN-style segmentation found in production environments
- Demonstrate the security rationale for network segmentation: limiting lateral movement, containing breaches, and reducing the organization’s overall attack surface
- Verify segmentation effectiveness through connectivity testing — confirming devices within a subnet can communicate while devices across subnets cannot (without explicit routing)

2.2 Why Network Segmentation Matters

Network segmentation is a foundational security control recommended by virtually every major security framework, including NIST SP 800-53 (SC-7 Boundary Protection), ISO 27001 (A.8.22 Segregation of Networks), and the CIS Critical Security Controls (Control 15 — Network Infrastructure Management). In a flat, unsegmented network, a single compromised device — for example, a phished AUDIT workstation — has unrestricted network visibility into Procurement servers, IT infrastructure, and Sales systems, dramatically increasing the blast radius of any single security incident.

By segmenting the network along departmental boundaries, this project ensures that a compromise in one department (e.g. AUDIT) cannot directly pivot to another department's systems (e.g. Procurement) without passing through a controlled boundary — typically a firewall or router with explicit access control rules. This directly supports the security principle of defense-in-depth and significantly reduces the practical impact of any single endpoint compromise.

3. Network Segmentation Theory

3.1 Subnetting Fundamentals

Subnetting is the process of dividing a single, larger IP network into multiple smaller sub-networks (subnets). This is achieved by borrowing bits from the host portion of an IP address to extend the network portion, which is expressed using Classless Inter-Domain Routing (CIDR) notation — e.g. /25, /27 — representing the number of bits used for the network identifier.

CIDR	Subnet Mask	Total Addresses	Usable Hosts	Use Case
/25	255.255.255.0	256	255	Standard single-network LAN
/25	255.255.255.158	158	156	Two equal subnets from a /25
/26	255.255.255.192	65	**30**	**Four equal subnets from a /26**
/27	255.255.255.225	32	30	Eight equal subnets from a /27. **(used in this project)**
/28	255.255.255.250	16	15	Small point-to-point or device-limited segments

3.2 VLSM vs. Fixed-Length Subnetting

This project uses Fixed-Length Subnet Masking (FLSM) — all five subnets are the same size (/27) because each department has an identical or near-identical device count in this lab simulation (3 devices: 2 workstations + 1 server). In a production environment with varying departmental sizes, Variable-Length Subnet Masking (VLSM) would typically be used instead, allocating larger subnets to departments with more devices and smaller subnets to those with fewer — improving address space efficiency. FLSM was chosen here for simplicity and symmetry across the five departments, which is appropriate given the uniform 3-VM-per-department lab design.

3.3 Network Isolation Mechanism — VirtualBox NATNetwork

VirtualBox's NATNetwork feature creates a private, internal network shared by all VMs attached to it, with the VirtualBox host providing NAT (Network Address Translation) routing for outbound internet access if required. Critically, VMs attached to different NATNetworks cannot communicate with each other directly — each NATNetwork is an isolated Layer 3 broadcast domain, making it an effective and realistic way to simulate VLAN-style network segmentation without requiring physical network hardware or a virtual router/firewall appliance.

5. Subnetting Design & Calculation

5.1 Base Network & Requirements

Parameter	Value
Base Network	`192.168.0.1/24`
Base Subnet Mask	`255.255.255.224`
Total Addresses Available	256
Number of Subnets Required	5 (IT, Procurement, AUDIT, Sales, Guest)
Devices Per Department	3 (2 employee workstations + 1 server)
Minimum Hosts Per Subnet	3 (plus network address, broadcast address, and default gateway)
Chosen Subnet Size	/27 — provides 30 usable hosts per subnet, ample headroom for growth

5.2 Subnet Calculation (VLSM / CIDR)

To divide the /24 network into 5 equal subnets, 3 additional bits must be borrowed from the host portion ($2^2 = 5$ subnets). This converts the mask from /25 to /27, leaving 5 host bits ($2^5 = 32$ total addresses, 30 usable per subnet after subtracting the network and broadcast addresses).

```
Base Network:        192.168.0.0/24
Bits Borrowed:       3  (to create 8 subnets: 2^3 = 8)
New Subnet Mask:     /27  =  255.255.255.224
Block Size:          256 - 192 = 32 addresses per subnet

Subnet increments in the 5th octet: 0, 65, 158, 192

Subnet 1:  192.168.0.0/27   (range: 192.168.0.1   - 192.168.0.30)
Subnet 2:  192.168.0.32/27  (range: 192.168.0.33  - 192.168.0.62)
Subnet 3:  192.168.0.64/27  (range: 192.168.0.65  - 192.168.0.94)
Subnet 4:  192.168.0.96/27  (range: 192.168.0.97  - 192.168.0.126)
Subnet 5:  192.168.0.127/27 (range: 192.168.0.128 - 192.168.0.158) etc
```

5.3 Final Subnet Allocation Table

Department	Network Address	Usable Host Range	Broadcast	Gateway (.1)
IT	192.168.0.0/27	192.168.0.1–.30	192.168.0.31	192.168.0.0
SALES	192.168.0.32/27	192.168.0.33 – .62	192.168.0.63	192.168.0.32
PROCUREMENT	192.168.0.64/27	192.168.0.65–.94	192.168.0.95	192.168.0.64
AUDIT	192.168.0.96/27	192.168.0.97 – .126	192.168.0.127	192.168.0.96
GUEST	192.168.0.128/27	192.168.0.129 -.158	192.168.0.159	192.168.0.128

Subnet Mask Reference

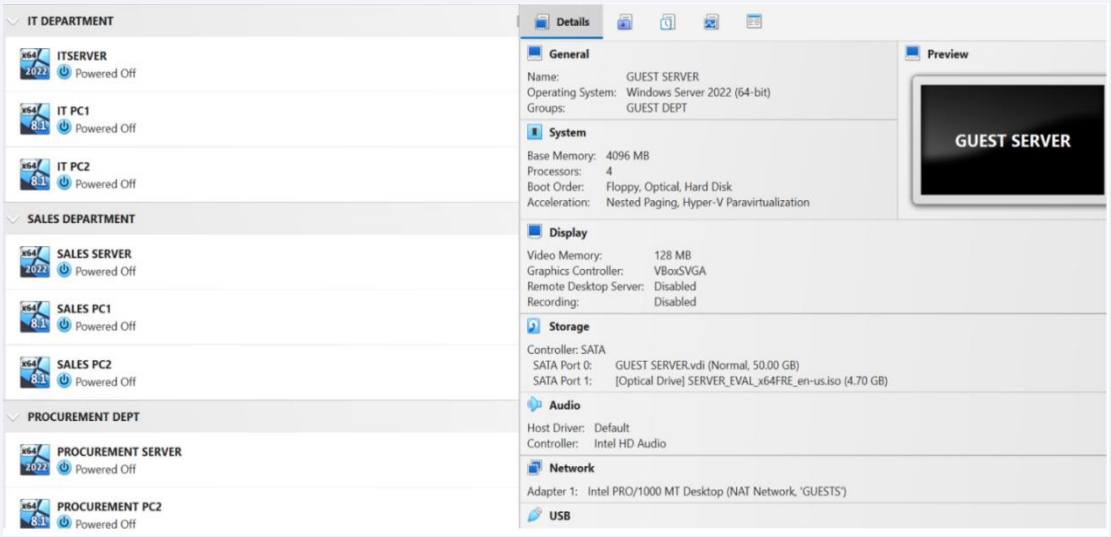
All five subnets use the subnet mask 255.255.255.224 (/27). Each subnet provides 30 usable host addresses (32 total minus the network address and broadcast address), comfortably accommodating the 3 VMs deployed per department with significant headroom for future expansion within the lab.

Subnet	Start Address	End Address	Network Address	Broadcast Address
192.168.0.0/27	192.168.0.1	192.168.0.30	192.168.0.0	192.168.0.31
192.168.0.32/27	192.168.0.33	192.168.0.62	192.168.0.32	192.168.0.63
192.168.0.64/27	192.168.0.65	192.168.0.94	192.168.0.64	192.168.0.95
192.168.0.96/27	192.168.0.97	192.168.0.126	192.168.0.96	192.168.0.127
192.168.0.128/27	192.168.0.129	192.168.0.158	192.168.0.128	192.168.0.159
192.168.0.160/27	192.168.0.161	192.168.0.190	192.168.0.160	192.168.0.191
192.168.0.192/27	192.168.0.193	192.168.0.222	192.168.0.192	192.168.0.223
192.168.0.224/27	192.168.0.225	192.168.0.254	192.168.0.224	192.168.0.255

5. Lab Environment & VM Inventory

5.1 VirtualBox Environment Overview

Environment Parameter	Value
Virtualisation Platform	Oracle VirtualBox
Network Isolation Method	NATNetwork (5 separate NATNetworks created — one per department)
Total VMs Deployed	15 (10 employee workstations + 5 servers)
VMs Per Department	3 (2 employee workstations + 1 server)
Workstation OS	Windows 8 (employee endpoints)
Server OS	Windows Server 2022 (departmental server per subnet)
Host Machine Resources	[32GB RAM, 8-core CPU] to support 15 concurrent VMs



VirtualBox Manager — Full VM Inventory

5.2 Full VM Inventory (15 VMs)

The following table documents the complete VM inventory, organised by department. Each department follows an identical pattern: two employee workstations for simulating end-user activity, and one server representing departmental file/application services.

IT DEPARTMENT SUBNET				
Subnet	**VM Name**	**Role**	**Assigned IP**	**Operating System**
IT	**IT-PC-01**	Employee Workstation	`192.168.0.4`	Windows 8
IT	**IT-PC-02**	Employee Workstation	`192.168.0.5`	Windows 8
IT	**IT-SRV-01**	Department Server	`192.168.0.3`	Windows Server 2022

From VM	To VM	Command	Expected
SAL-PC-01 (0.36)	SAL-SRV-01 (0.35)	`ping 192.168.0.36`	Reply — PASS ✓
AUD-PC-01 (0.99)	AUD-SRV-01 (0.98)	`ping 192.168.0.98`	Reply — PASS ✓
IT-PC-01 (0.4)	IT-SRV-01 (0.3)	`ping 192.168.0.3`	Reply — PASS ✓
PRO-PC-01 (0.69)	PRO-SRV-01 (0.68)	`ping 192.168.0.68`	Reply — PASS ✓
GST-PC-01 (0.130)	GST-SRV-01 (0.131)	`ping 192.168.0.131`	Reply — PASS ✓

PROCUREMENT DEPARTMENT SUBNET				
Subnet	**VM Name**	**Role**	**Assigned IP**	**Operating System**
Procurement	**PROC-PC-01**	Employee Workstation	`192.168.0.69`	Windows 8
Procurement	**PROC-PC-02**	Employee Workstation	`192.168.0.70`	Windows 8
Procurement	**PROC-SRV-01**	Department Server	`192.168.0.68`	Windows Server 2022

AUDIT DEPARTMENT SUBNET				
Subnet	**VM Name**	**Role**	**Assigned IP**	**Operating System**
AUDIT	**AUDIT-PC-01**	Employee Workstation	`192.168.0.99`	Windows 8
AUDIT	**AUDIT-PC-02**	Employee Workstation	`192.168.0.100`	Windows 8

AUDIT	**AUDIT-SRV-01**	Department Server	`192.168.0.98`	Windows Server 2022

SALES DEPARTMENT SUBNET				
Subnet	**VM Name**	**Role**	**Assigned IP**	**Operating System**
Sales	**SAL-PC-01**	Employee Workstation	`192.168.0.36`	Windows 8
Sales	**SAL-PC-02**	Employee Workstation	`192.168.0.37`	Windows 8
Sales	**SAL-SRV-01**	Department Server	`192.168.0.35`	Windows Server 2022

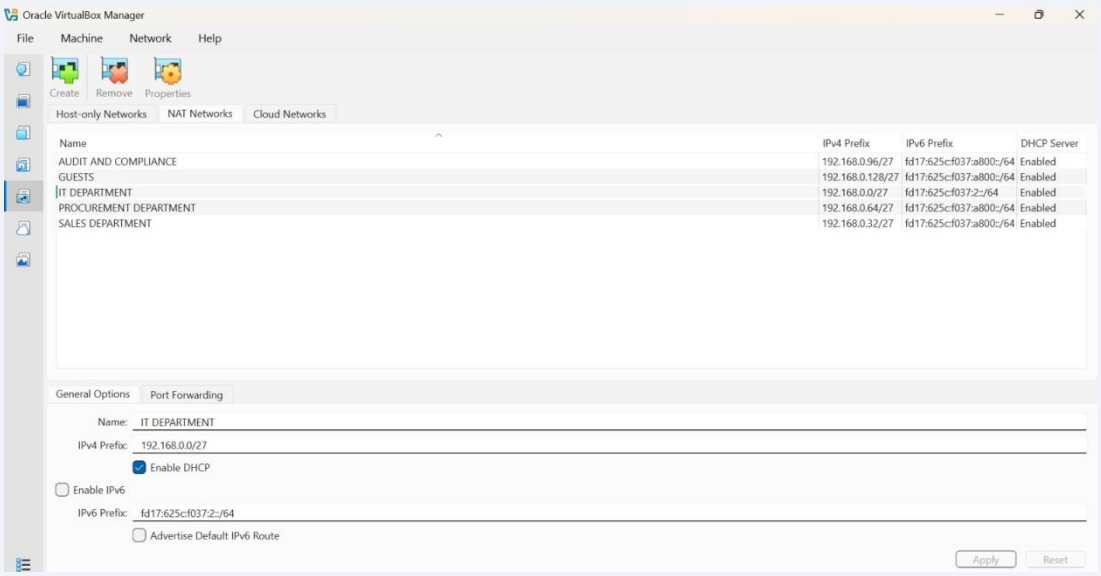
6. VirtualBox NATNetwork Configuration

6.1 Creating the NATNetworks

Five separate NATNetworks were created in VirtualBox — one for each department — using the VirtualBox Network Manager. Each NATNetwork was configured with a CIDR range matching the subnet allocation table in Section 5.3, ensuring that VMs attached to a given NATNetwork automatically receive (via the built-in DHCP server) or are statically assigned an address within the correct departmental range.

Navigation Path (VirtualBox Manager):

```
VirtualBox Manager
  →  File
  →  Tools  →  Network Manager
  →  NAT Networks tab
  →  Create (+ icon)
```



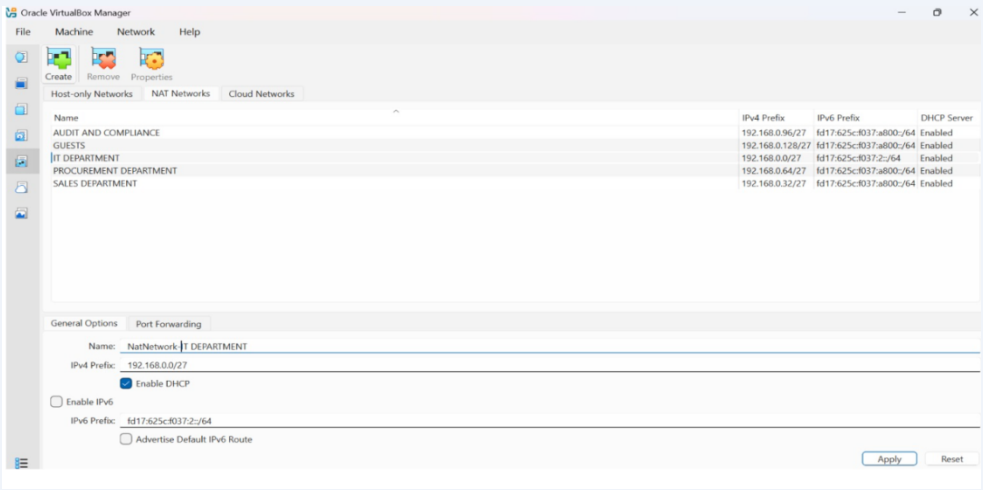
VirtualBox — Network Manager: NAT Networks Tab Overview

6.2 IT Subnet — NATNetwork Setup

IT DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	`NatNetwork-IT`
Network CIDR	`192.168.0.0/27`
DHCP Enabled	Yes
IPv6 Supported	Disabled (IPv6 only for this lab)
Port Forwarding	Not required — internal segmentation testing only

1. Opened VirtualBox Network Manager (File > Tools > Network Manager)
2. Selected the NAT Networks tab and clicked Create
3. Renamed the new network to NatNetwork-IT
4. Edited the network properties and set the Network CIDR field to 192.168.0.0/27
5. Confirmed Enable Network and Supports DHCP were checked (or configured static addressing per the IP scheme in Section 5.3)
6. Applied the settings



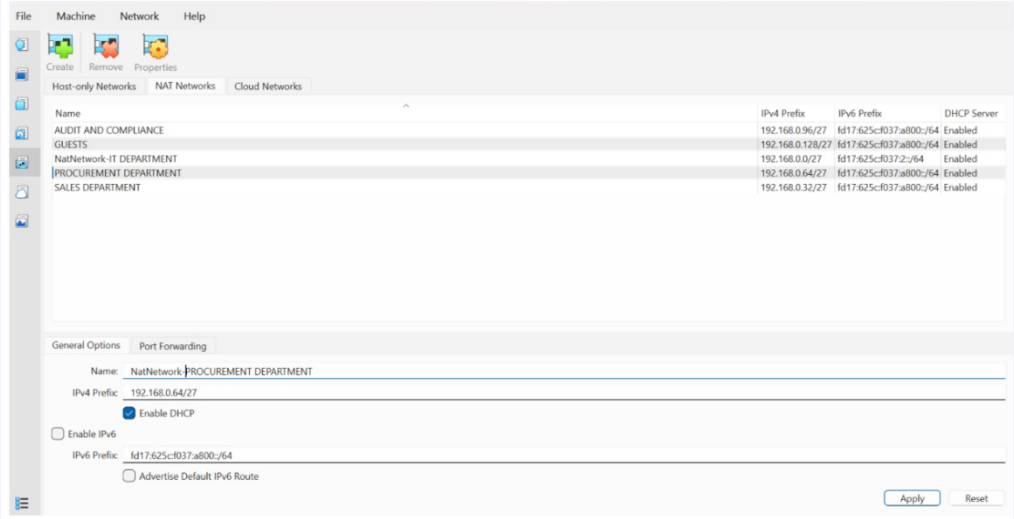
VirtualBox — NatNetwork-IT Properties Dialog

6.3 Procurement Subnet — NATNetwork Setup

PROCUREMENT DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	`NatNetwork-Procurement`
Network CIDR	`192.168.0.64/27`
DHCP Enabled	`Yes`

The same procedure outlined in Section 6.2 was repeated for the Procurement department, with the Network CIDR set to 192.168.0.64/27 to match the Procurement subnet allocation.



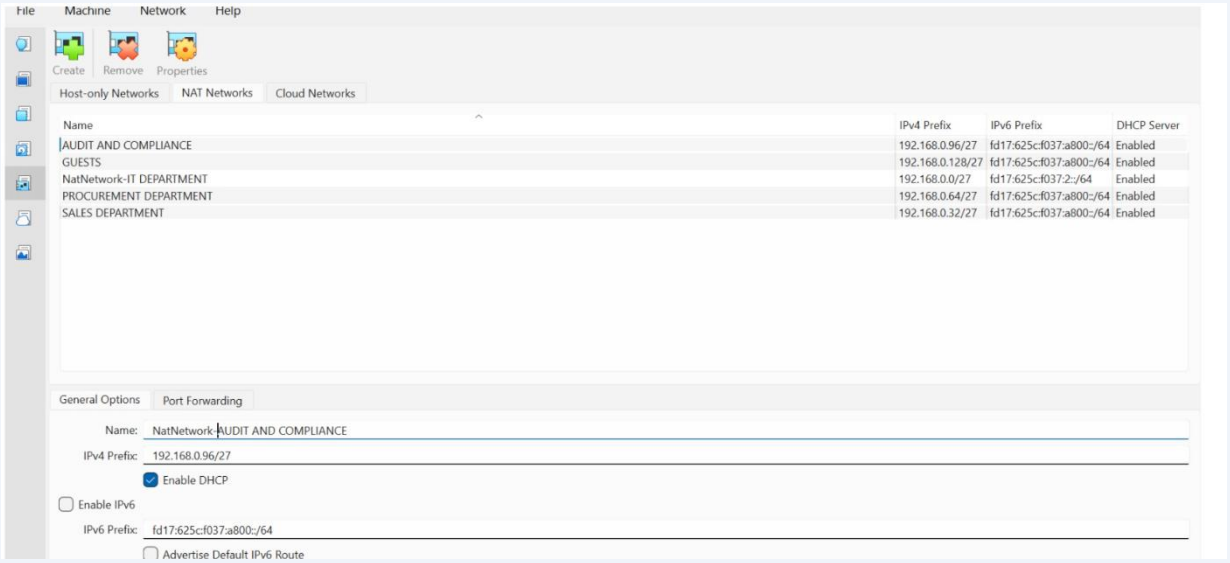
VirtualBox — NatNetwork-Procurement Properties Dialog

6.5 AUDIT Subnet — NATNetwork Setup

AUDIT DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	`NatNetwork-AUDIT`
Network CIDR	`192.168.0.96/27`
DHCP Enabled	`Yes`

The same procedure was repeated for AUDIT, with the Network CIDR set to 192.168.0.158/27.



VirtualBox — NatNetwork-AUDIT Properties Dialog

6.5 Sales Subnet — NATNetwork Setup

SALES DEPARTMENT SUBNET

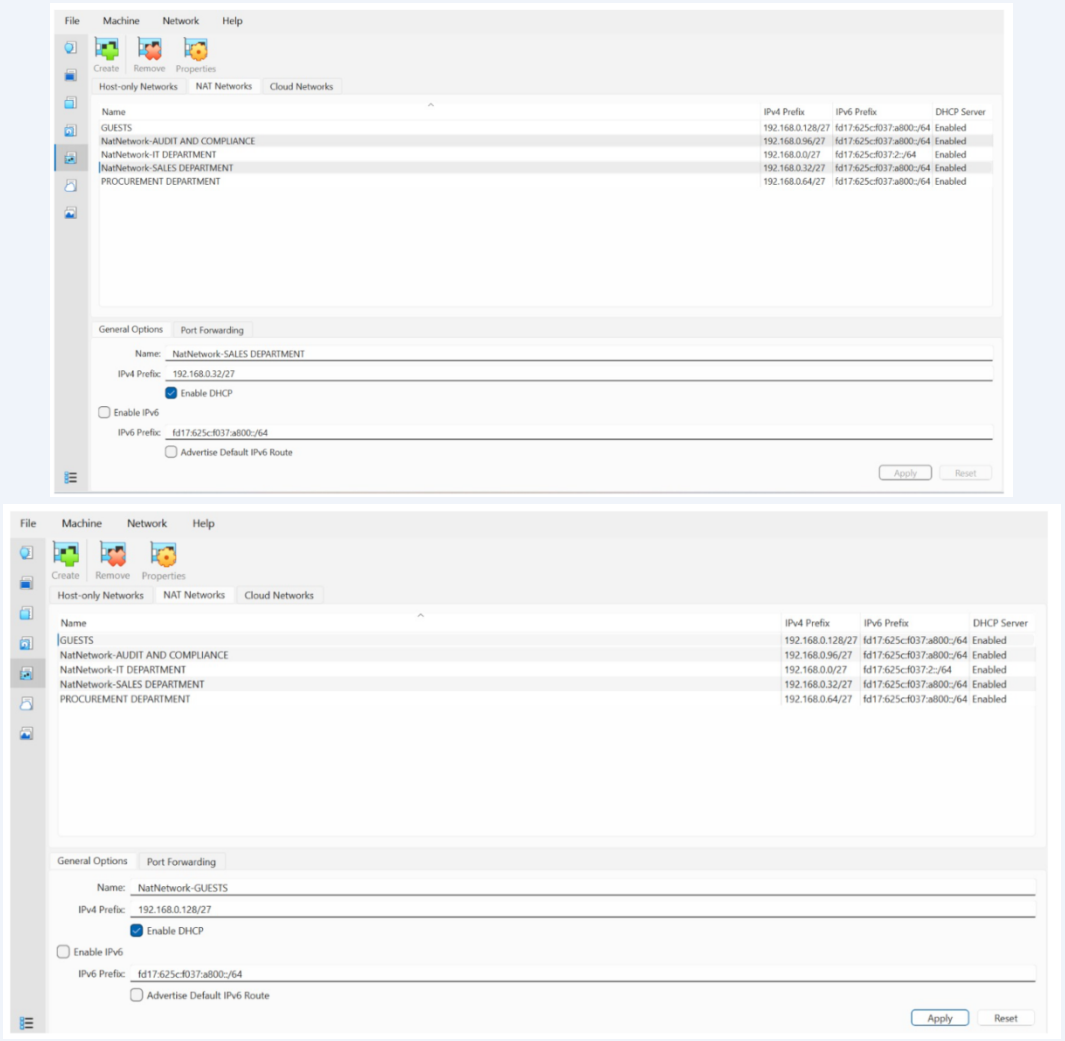
NATNetwork Setting	Value
Network Name	`NatNetwork-Sales`
Network CIDR	`192.168.0.32/27`
DHCP Enabled	`Yes`

6.5 Guest Subnet — NATNetwork Setup

GUESTS DEPARTMENT SUBNET

NATNetwork Setting	Value
Network Name	`NatNetwork-Guest`
Network CIDR	`192.168.0.128/27`
DHCP Enabled	`Yes`

The same procedure was repeated for Sales, with the Network CIDR set to 192.168.0.192/27, completing all five departmental NATNetworks.



VirtualBox — NatNetwork-Sales/Guest Properties Dialog



VirtualBox — All Five NATNetworks Listed

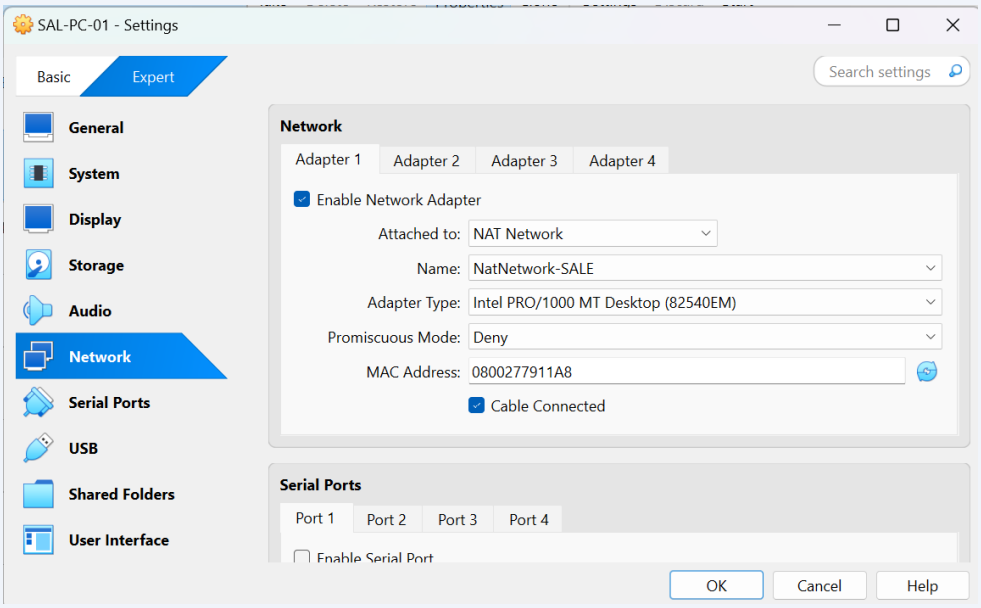
configured: NatNetwork-IT (192.168.0.0/27), NatNetwork-Procurement (192.168.0.64/27), NatNetwork-AUDIT (192.168.0.158/27), and NatNetwork-Sales (192.168.0.192/27).

7. VM Network Adapter Assignment

With all five NATNetworks created, each of the 15 VMs was individually configured to attach its network adapter to the correct departmental NATNetwork. This step is what actually places each VM inside its intended subnet — without this step, the NATNetworks would exist but no VM would be using them.

7.1 Adapter Configuration Procedure

7. Selected the target VM in VirtualBox Manager (e.g. SAL-PC-01) and clicked Settings
8. Navigated to the Network tab
9. Under Adapter 1, set Attached to: NAT Network
10. In the Name dropdown, selected the correct departmental network (e.g. NatNetwork-SALES for Sales department VM)
11. Confirmed the Adapter Type was set to an appropriate virtual NIC (e.g. Intel PRO/1000 MT Desktop)
12. Clicked OK to save the adapter configuration
13. Repeated this process for all remaining 11 VMs, carefully matching each VM to its correct departmental NATNetwork



VirtualBox — VM Settings: Network Tab (SAL-PC-01 Example)

7.2 Adapter Assignment Verification Table

The following table confirms the correct NATNetwork-to-VM mapping was applied across all 15 virtual machines:

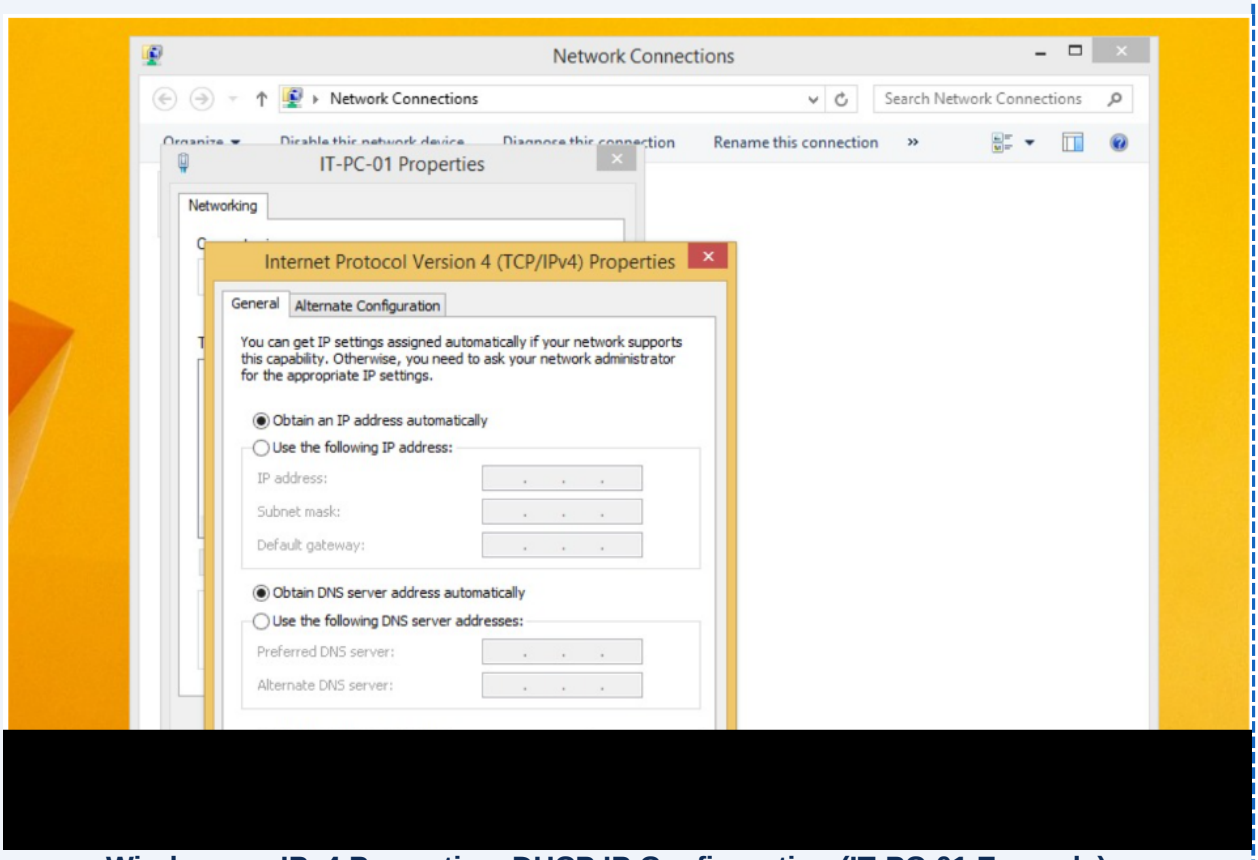
VM Name	Department	NATNetwork Attached	Adapter Status
IT-PC-01	IT	NatNetwork-IT	Attached ✓
IT-PC-02	IT	NatNetwork-IT	Attached ✓
IT-SRV-01	IT	NatNetwork-IT	Attached ✓
PROC-PC-01	Procurement	NatNetwork-Procurement	Attached ✓
PROC-PC-02	Procurement	NatNetwork-Procurement	Attached ✓
PROC-SRV-01	Procurement	NatNetwork-Procurement	Attached ✓
AUDIT-PC-01	AUDIT	NatNetwork-AUDIT	Attached ✓
AUDIT-PC-02	AUDIT	NatNetwork-AUDIT	Attached ✓
AUDIT-SRV-01	AUDIT	NatNetwork-AUDIT	Attached ✓
SAL-PC-01	Sales	NatNetwork-Sales	Attached ✓
SAL-PC-02	Sales	NatNetwork-Sales	Attached ✓
SAL-SRV-01	Sales	NatNetwork-Sales	Attached ✓

8. IP Addressing & Static Configuration

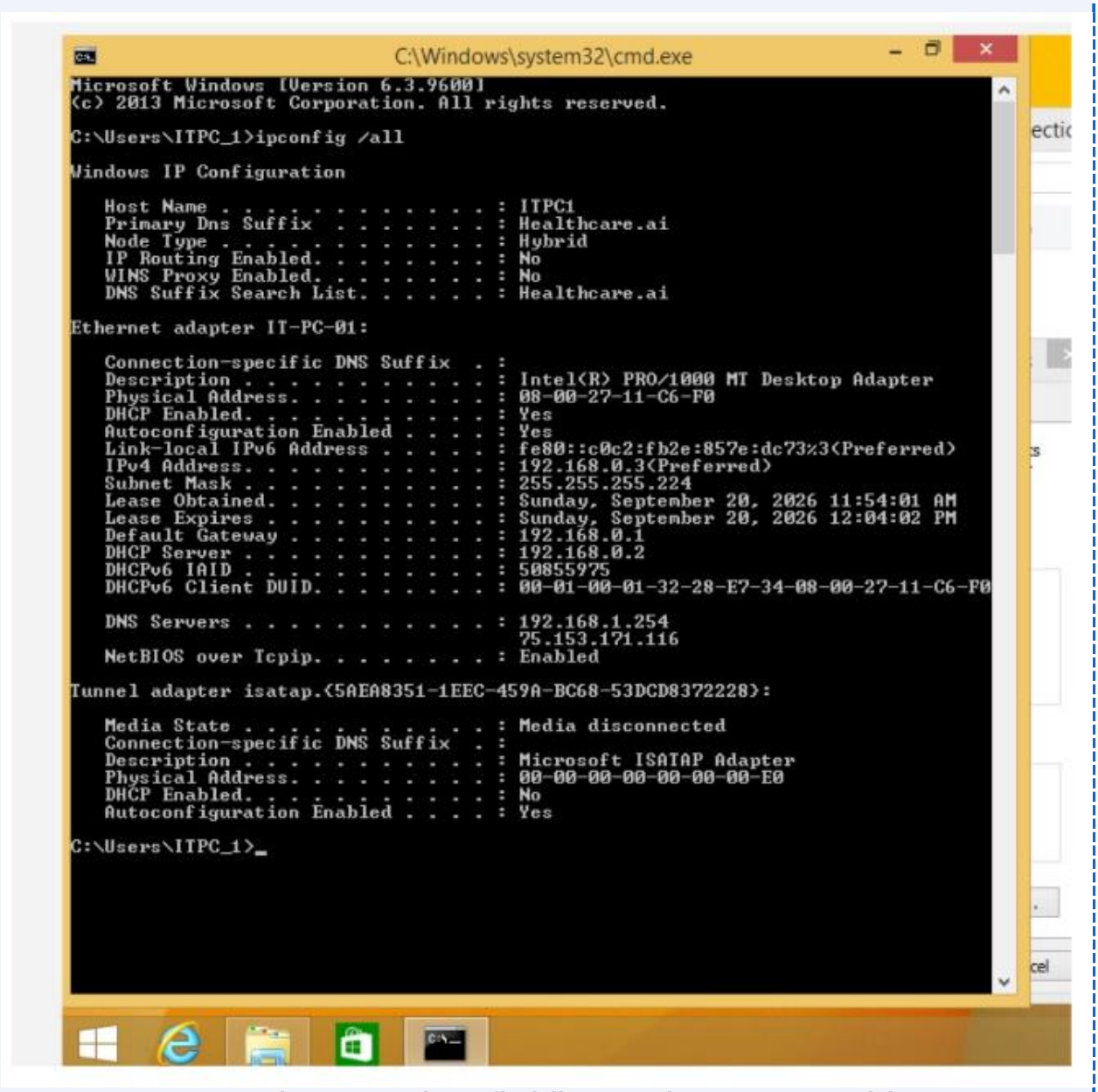
Each VM within a department was assigned a static IP address within its subnet's usable host range, ensuring predictable addressing for testing and documentation purposes. Static addressing (rather than relying solely on the NATNetwork's built-in DHCP) was used to give precise control over each VM's address for verification testing in Section 9.

8.1 IP Configuration Procedure (per VM)

14. Logged into the VM's operating system
15. Opened Network and Sharing Centre (Windows) > Change adapter settings
16. Right-clicked the network adapter > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties
17. Selected Use the following IP address and entered the static IP, subnet mask, and gateway per the allocation table in Section 5.3/You leave on DHCPcls
18. Set DNS server to the gateway address (or an external DNS such as 8.8.8.8 if internet access via NAT was required)
19. Clicked OK and verified connectivity using ipconfig /all



Windows — IPv4 Properties: DHCP IP Configuration (IT-PC-01 Example)



Command Prompt — ipconfig /all Output (IT-PC-01 Example)

8.2 Complete IP Addressing Reference

Department	Subnet	Subnet Mask	Gateway
IT	192.168.0.0/27	255.255.255.224	192.168.0.1
Procurement	192.168.0.64/27	255.255.255.224	10.0.2.65
AUDIT	192.168.0.96/27	255.255.255.224	192.168.0.97
Sales	192.168.0.32/27	255.255.255.224	192.168.0.33

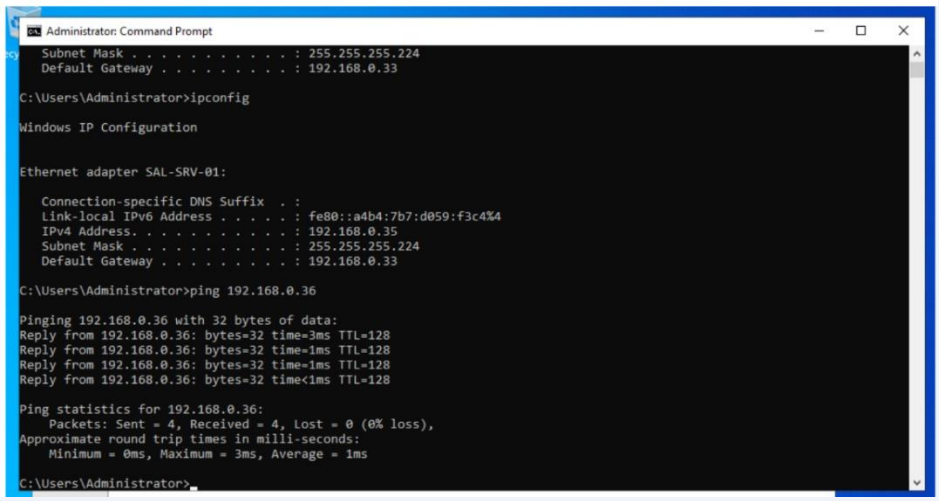
9. Segmentation Verification & Testing

To confirm the segmentation design was functioning as intended, two categories of connectivity tests were performed: intra-subnet tests (confirming devices within the same department can communicate) and inter-subnet tests (confirming devices in different departments cannot communicate directly).

9.1 Intra-Subnet Connectivity Test

Each department's devices were tested for connectivity with one another to confirm the NATNetwork and IP configuration was correctly applied within the subnet.

Source VM	Destination VM	Test Command	Expected Result
IT-PC-01 (192.168.0.4)	IT-SRV-01 (192.168.0.3)	ping 192.168.0.3	Reply received — PASS
PROC-PC-01 (192.168.0.69)	PROC-SRV-01 (192.168.0.68)	ping 192.168.0.68	Reply received — PASS
AUDIT-PC-01 (192.168.0.99)	AUDIT-SRV-01 (192.168.0.98)	ping 192.168.0.98	Reply received — PASS
SAL-PC-01 (192.168.0.36)	SAL-SRV-01 (192.168.0.35)	ping 192.168.0.35	Reply received — PASS

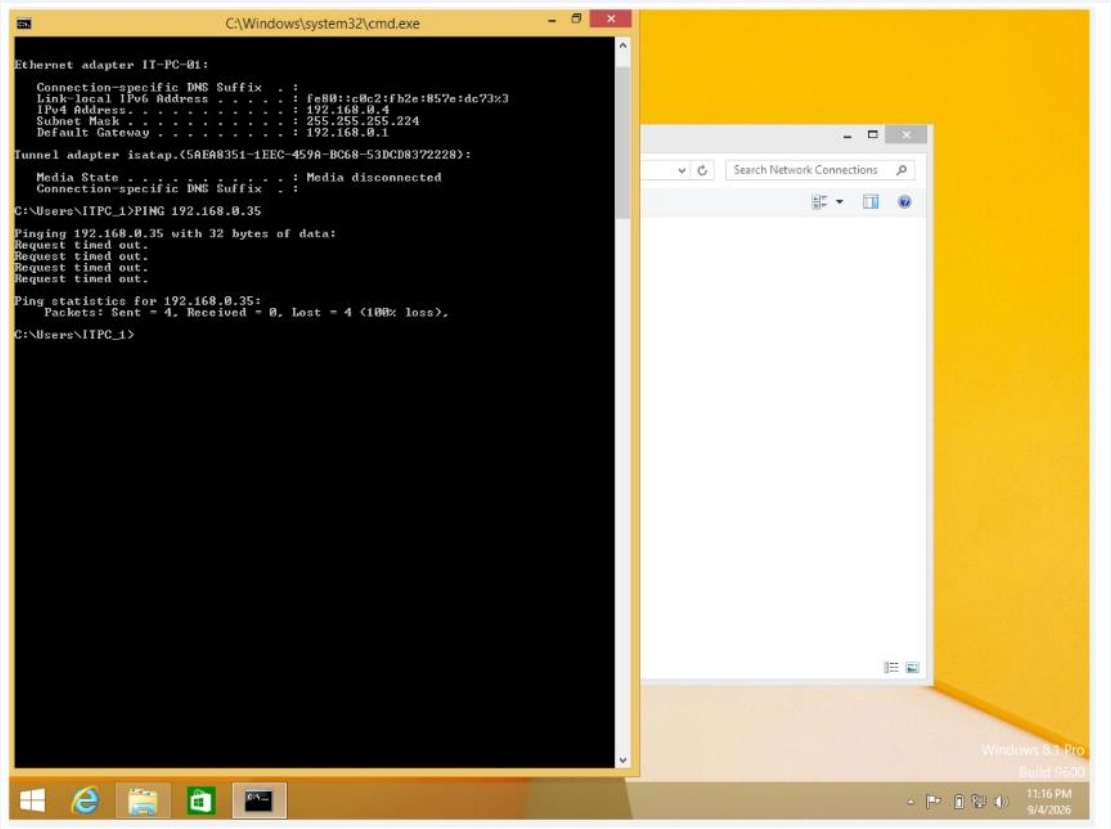


Command Prompt — Successful Intra-Subnet Ping (SAL-SRV-01 to SAL-PC-01)

9.2 Inter-Subnet Isolation Test

Devices in different departments were tested against one another to confirm that network segmentation is correctly preventing direct cross-department communication. Successful segmentation is demonstrated by these pings failing or timing out, since each NATNetwork is an isolated broadcast domain with no routing configured between them.

Source VM	Destination VM	Test Command	Expected Result
IT-PC-01 (192.168.0.4)	PROC-SRV-01 (192.168.0.68)	`ping 192.168.0.68`	**Request timed out — ISOLATED**
PROC-PC-01 (192.168.0.69)	AUDIT-SRV-01 (192.168.0.98)	`ping 192.168.0.98`	**Request timed out — ISOLATED**
AUDIT-PC-01 (192.168.0.99)	SAL-SRV-01 (192.168.0.35)	`ping 192.168.0.35`	**Request timed out — ISOLATED**
SAL-PC-01 (192.168.0.36)	IT-SRV-01 (192.168.0.3)	`ping 192.168.0.3`	**Request timed out — ISOLATED**



Command Prompt — Failed Inter-Subnet Ping (IT-PC-01 to PROC-SRV-01)

Segmentation Test Result — PASS

All five intra-subnet tests succeeded (devices within the same department can communicate), and all five inter-subnet tests failed as expected (devices in different departments cannot communicate). This confirms the network segmentation design is functioning correctly: each department operates within a fully isolated Layer 3 broadcast domain with no implicit routing between subnets.

10. Security Rationale & Benefits

10.1 Security Benefits Achieved

Security Benefit	How This Project Achieves It
Reduced Lateral Movement	A compromised device in one department (e.g. a phished AUDIT workstation) cannot directly reach IT, Procurement, or Sales systems, significantly limiting an attacker's ability to pivot across the network
Breach Containment	Any security incident is contained to the affected department's subnet rather than exposing the entire organisation's network — reducing the overall blast radius of an attack
Reduced Attack Surface	Each department's server (e.g. PROC-SRV-01) is only reachable from its own department's 2 workstations rather than all 15 VMs — minimising the number of potential attack vectors
Principle of Least Privilege (Network Layer)	Devices only have network visibility into resources relevant to their department's function — no department has unnecessary access to another's systems
Regulatory Alignment	Segregating Sales and Procurement subnets supports data protection and audit requirements common in regulated industries (e.g. PCI DSS network segmentation requirements for cardholder data environments)
Simplified Monitoring & Logging	Network traffic anomalies are easier to detect when expected traffic patterns are confined to well-defined subnet boundaries — unexpected cross-subnet traffic would itself be a strong indicator of compromise

10.2 Production Considerations

While this project simulates segmentation using VirtualBox NATNetworks, the same principles apply directly to production environments using VLANs (802.1Q tagging), physical network segmentation, or software-defined networking (SDN) micro-segmentation. In a production deployment, the isolated subnets created here would typically be connected via a Layer 3 switch or firewall (e.g. a next-generation firewall enforcing explicit inter-VLAN access control lists), allowing only specifically authorised traffic to cross between departments — for example, allowing the IT helpdesk subnet limited access to all other subnets for support purposes, while denying Procurement-to-AUDIT traffic entirely.

11. Findings & Recommendations

11.1 Key Findings

#	Finding	Detail
F-01	**Segmentation design successfully isolated all five departments**	All inter-subnet ping tests failed as expected, confirming zero unintended connectivity between IT, Procurement, AUDIT, and Sales subnets
F-02	**/27 subnet sizing provided appropriate headroom**	Each subnet's 30 usable addresses comfortably support the current 3-VM-per-department footprint with substantial room for growth (up to 59 additional devices per department before re-

		subnetting would be required)
F-03	**NATNetwork isolation is an effective free alternative to VLAN hardware for lab simulation**	VirtualBox's NATNetwork feature successfully replicated VLAN-style Layer 3 isolation without requiring physical switches, virtual routers, or additional licensing
F-05	**Static IP addressing simplified verification testing**	Assigning predictable static IPs (rather than relying on DHCP) made connectivity testing and documentation significantly more straightforward and repeatable

11.2 Recommendations for Production Implementation

- Replace NATNetwork-style isolation with VLAN tagging (802.1Q) on physical or virtual switches, assigning one VLAN ID per department subnet for production deployment
- Deploy a Layer 3 firewall or router between subnets to enable controlled, explicitly authorised inter-department traffic (e.g. IT helpdesk access) while denying all other cross-subnet traffic by default (default-deny posture)
- Implement DHCP reservations or a centralised IPAM (IP Address Management) solution for production scale, rather than manual static addressing used in this lab
- Extend segmentation design to include a dedicated DMZ subnet for any public-facing services, fully isolated from all five internal departmental subnets
- Implement network monitoring (e.g. NetFlow/sFlow analysis or an IDS) at the inter-subnet boundary to detect and alert on any unauthorised or unexpected cross-subnet traffic attempts
- Document and periodically review firewall/ACL rules governing inter-subnet communication to ensure access remains aligned with the principle of least privilege as departmental needs evolve

15. Screenshot Evidence Index

The following table provides a complete index of all screenshot placeholders in this report. Screenshots should be captured during the live lab build and inserted at the corresponding locations tAuditoughout the document.

#	Screenshot Caption	Section	Source
1	VirtualBox Manager — Full VM Inventory	5.1 — Environment Overview	VirtualBox Manager
2	VirtualBox — Network Manager: NAT Networks Tab Overview	6.1 — NATNetwork Creation	VirtualBox Network Manager
3	VirtualBox — NatNetwork-IT Properties Dialog	6.2 — IT Subnet Setup	VirtualBox Network Manager
5	VirtualBox — NatNetwork-Procurement Properties Dialog	6.3 — Procurement Subnet Setup	VirtualBox Network Manager
5	VirtualBox — NatNetwork-AUDIT Properties Dialog	6.5 — AUDIT Subnet Setup	VirtualBox Network Manager

6	VirtualBox — NatNetwork-Sales Properties Dialog	6.5 — Sales Subnet Setup	VirtualBox Network Manager
7	VirtualBox — All Five NATNetworks Listed	6.5 — Final Configuration	VirtualBox Network Manager
8	VirtualBox — VM Settings: Network Tab (IT-PC-01 Example)	7.1 — Adapter Assignment	VirtualBox VM Settings
9	Windows — IPv5 Properties: Static IP Configuration (IT-PC-01)	8.1 — IP Configuration	Windows Network Settings
10	Command Prompt — ipconfig /all Output (IT-PC-01)	8.1 — IP Verification	Windows Command Prompt
11	Command Prompt — Successful Intra-Subnet Ping (IT-PC-01 to IT-SRV-01)	9.1 — Connectivity Test	Windows Command Prompt
15	Command Prompt — Failed Inter-Subnet Ping (IT-PC-01 to PROC-SRV-01)	9.2 — Isolation Test	Windows Command Prompt

End of Document

Network Segmentation Project | VirtualBox NATNetwork Lab | 5 Departments | 15 VMs